Antiderivatives, bounds and small elements

The definite integral

If f is continuous on $[a, b]$ and $F' = f$, then

$$\int_a^b f(t)\,\mathrm{d}t = F(b) - F(a).$$

The variable t describes what changes. The bounds specify its starting and ending values. A definite integral is a number; the integration variable disappears after evaluation.
Useful antiderivatives are

Integrand	An antiderivative
t^n $(n \neq -1)$	$t^{n+1}/(n+1)$
$1/t$ $(t \neq 0)$	$\ln\lvert t\rvert$
e^{kt} $(k \neq 0)$	e^{kt}/k
$\cos t$	$\sin t$
$\sin t$	$-\cos t$

For a substitution $u = g(t)$, replace the differential and both bounds:

$$\int_a^b F(g(t))g'(t)\,\mathrm{d}t = \int_{g(a)}^{g(b)} F(u)\,\mathrm{d}u.$$

For example, with $u = s^2 + s + 1$, we have $\mathrm{d}u = (2s+1)\,\mathrm{d}s$. The bounds $s = 0, 3$ become $u = 1, 13$, so

$$\int_0^3 (2s+1)\sqrt{s^2+s+1}\,\mathrm{d}s = \int_1^{13} u^{1/2}\,\mathrm{d}u = \frac{2}{3}\left(13\sqrt{13} - 1\right).$$

Do not combine an antiderivative in u with bounds still expressed as values of s.

What is being added?

Quantity	Small contribution	Typical units
Curve length	$\mathrm{d}s$	length
Curved surface area	$2\pi(\text{radius})\,\mathrm{d}s$	length^2
Volume by slices	$(\text{cross-sectional area})\,\mathrm{d}x$	length^3
Planar area	$\mathrm{d}A$	length^2
Volume below a surface	$(\text{height})\,\mathrm{d}A$	length^3
Volume by small boxes	$\mathrm{d}V$	length^3

The integrand and differential must describe the same small piece. In an inner integral, hold the other variables constant; for example, $\int 3xy^2\,\mathrm{d}y = xy^3 + C$ when x is fixed.

Cylinders, quadrics and traces

Recognising the surface

A *trace* is the curve obtained by fixing one coordinate. If a surface equation does not involve one coordinate, its cross-section repeats parallel to that coordinate axis: the surface is a cylinder.
For positive a, b, c, these standard equations describe the six quadric families. Interchanging coordinates changes the direction of the axis.

Equation	Surface and a useful observation
$\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$	Ellipsoid; all coordinates are bounded.
$z=\frac{x^2}{a^2}+\frac{y^2}{b^2}$	Elliptic paraboloid; $z=k>0$ gives ellipses.
$z=\frac{x^2}{a^2}-\frac{y^2}{b^2}$	Hyperbolic paraboloid; the two vertical central traces open in opposite directions.
$\frac{x^2}{a^2}+\frac{y^2}{b^2}-\frac{z^2}{c^2}=1$	One-sheet hyperboloid; every horizontal level has an elliptical trace.
$\frac{z^2}{c^2}-\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$	Two-sheet hyperboloid; no points satisfy $\lvert z\rvert<c$.
$\frac{x^2}{a^2}+\frac{y^2}{b^2}=\frac{z^2}{c^2}$	Elliptic cone; the trace $z=0$ is a single point.

Complete squares before identifying a translated surface. An equation that requires a sum of nonnegative squares to equal a negative number has no real solutions.

Example 1. Identify the surface before choosing bounds

Identify and describe

$$x^2+4y^2-2x+8y-z+1=0.$$

Find its lowest point, its trace in $z=0$, and its traces in the two vertical planes through its axis. Also identify the surface $y^2+z^2-4y+2z=4$.

Solution. Complete the squares without changing the equation:

$$(x-1)^2-1+4\big((y+1)^2-1\big)-z+1=0,$$

so

$$\boxed{z=(x-1)^2+4(y+1)^2-4.}$$

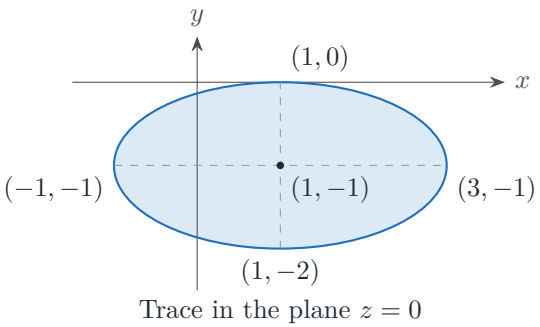
Both squares are nonnegative. Their smallest simultaneous value is zero at $x=1, y=-1$. Thus this is an elliptic paraboloid opening upwards, with vertex $(1,-1,-4)$ and axis $x=1, y=-1$.

At $z=0$,

$$(x-1)^2+4(y+1)^2=4 \quad\Longleftrightarrow\quad \frac{(x-1)^2}{4}+(y+1)^2=1.$$

This ellipse has centre $(1,-1)$, horizontal semiaxis 2 and vertical semiaxis 1. Its extreme x-values are $-1, 3$; its extreme y-values are $-2, 0$.

In the plane $y=-1$, the trace is $z=(x-1)^2-4$. In the plane $x=1$, it is $z=4(y+1)^2-4$. Both are upward-opening parabolas through the vertex; the second is narrower.



Trace in the plane $z=0$

For the second surface,

$$(y-2)^2+(z+1)^2=9.$$

There is no x in the equation. Every plane $x=k$ cuts out the same circle of radius 3, centred at $(y,z)=(2,-1)$. The surface is a circular cylinder parallel to the x-axis, with axis $y=2, z=-1$. It extends indefinitely in both x directions.

Arc length

The length element

A short smooth arc satisfies $\Delta s \approx \sqrt{(\Delta x)^2+(\Delta y)^2}$. Taking the limiting sum gives

$$y=f(x),\ a\le x\le b: \qquad L=\int_a^b \sqrt{1+[f'(x)]^2}\,\mathrm{d}x,$$

$$x=g(y),\ c\le y\le d: \qquad L=\int_c^d \sqrt{1+[g'(y)]^2}\,\mathrm{d}y.$$

These formulas apply when the derivative is continuous on the interval; they also apply piece by piece to suitable smooth arcs. Use the coordinate whose stated interval describes the whole required arc once. A derivative is squared, but a square root is nonnegative: $\sqrt{h^2}=|h|$.

Example 2. Simplify the length element first

Find the length of $y=\dfrac{1}{3}(x^2+2)^{3/2}$ from $x=0$ to $x=2$.

Solution. The curve is given as a smooth function of x, and the endpoints specify $0\le x\le 2$. Differentiate using the chain rule:

$$y'=\frac{1}{3}\cdot\frac{3}{2}(x^2+2)^{1/2}\cdot 2x=x\sqrt{x^2+2}.$$

Consequently,

$$1+(y')^2=1+x^2(x^2+2)=x^4+2x^2+1=(x^2+1)^2.$$

Since $x^2+1>0$, the length element is $(x^2+1)\,\mathrm{d}x$. Therefore

$$L=\int_0^2 (x^2+1)\,\mathrm{d}x=\left[\frac{x^3}{3}+x\right]_0^2=\boxed{\frac{14}{3}}.$$

The integral $\int y\,\mathrm{d}x$ would give area below the graph; it does not measure distance along it.

Example 3. Use the variable in which the curve is given

Find the length of

$$x=\frac{e^{2y}+e^{-2y}}{4}, \qquad 0\le y\le \ln 2.$$

Solution. There is no need to solve for y in terms of x. Use $\mathrm{d}s=\sqrt{1+(\mathrm{d}x/\mathrm{d}y)^2}\,\mathrm{d}y$ with the given y-bounds.

$$\frac{\mathrm{d}x}{\mathrm{d}y}=\frac{e^{2y}-e^{-2y}}{2}.$$

Because $e^{2y}e^{-2y}=1$,

$$1+\left(\frac{\mathrm{d}x}{\mathrm{d}y}\right)^2=\frac{e^{4y}+2+e^{-4y}}{4}=\left(\frac{e^{2y}+e^{-2y}}{2}\right)^2.$$

The expression in the final brackets is positive. Thus

$$L = \int_0^{\ln 2} \frac{e^{2y} + e^{-2y}}{2}\,\mathrm{d}y$$
$$= \left[\frac{e^{2y} - e^{-2y}}{4}\right]_0^{\ln 2} = \frac{4 - 1/4}{4} = \boxed{\frac{15}{16}}.$$

The first endpoint is $(1/2, 0)$ and the second is $(17/16, \ln 2)$. These are points on the curve, not alternative integration limits for an integral in y.

A curve described by a parameter

If $x = x(t), y = y(t)$ trace the required arc once for $\alpha \le t \le \beta$, then

$$L = \int_\alpha^\beta \sqrt{[x'(t)]^2 + [y'(t)]^2}\,\mathrm{d}t.$$

This follows by replacing $\mathrm{d}x$ and $\mathrm{d}y$ in $\mathrm{d}s^2 = \mathrm{d}x^2 + \mathrm{d}y^2$. A symmetry multiplier is valid only when the pieces have equal lengths and do not overlap except at endpoints.

Example 4. An astroid with unequal horizontal and vertical scales

Find the total length of the closed curve

$$x = 3\cos^3 t, \qquad y = 2\sin^3 t, \qquad 0 \le t \le 2\pi.$$

Solution. For $0 \le t \le \pi/2$, both coordinates are nonnegative. The curve travels from $(3, 0)$ to $(0, 2)$ along its first-quadrant arc. Reflections in the axes give four equal-length arcs, so calculate this arc and multiply by 4.

$$x' = -9\cos^2 t \sin t, \qquad y' = 6\sin^2 t \cos t.$$

On this interval, $\sin t \cos t \ge 0$. Hence

$$\sqrt{(x')^2 + (y')^2} = 3\sin t \cos t \sqrt{9\cos^2 t + 4\sin^2 t} = 3\sin t \cos t \sqrt{4 + 5\cos^2 t}.$$

Put $u = 4 + 5\cos^2 t$. Then $\mathrm{d}u = -10\sin t \cos t\,\mathrm{d}t$, and $t = 0, \pi/2$ give $u = 9, 4$. Reversing the new bounds absorbs the minus sign:

$$L_{\text{quarter}} = -\frac{3}{10}\int_9^4 u^{1/2}\,\mathrm{d}u = \frac{3}{10}\cdot\frac{2}{3}[u^{3/2}]_4^9$$
$$= \frac{1}{5}(27 - 8) = \frac{19}{5}.$$

Thus $\boxed{L = 76/5}$. The speed vanishes at the four tips, but each quadrant is a finite smooth arc away from its endpoints, so adding its length is valid.

Curved surface area of revolution

Radius times arc length

Rotating a short sloping curve segment produces a narrow band. Its area is approximately its circumference times its sloping length. Thus

$$\boxed{S = \int 2\pi\rho \, \mathrm{d}s},$$

where ρ is the perpendicular distance from the curve to the rotation axis. For the x-axis, $\rho = |y|$; for the y-axis, $\rho = |x|$. For a horizontal line $y = k$, $\rho = |y - k|$.
Use $\mathrm{d}s = \sqrt{1 + (y')^2}\,\mathrm{d}x$ or $\sqrt{1 + (x')^2}\,\mathrm{d}y$ as appropriate. The formula gives the *curved* surface swept out by the curve; it does not add end disks. Assume a smooth arc producing the surface without repeated coverage.

Example 5. Surface area about the horizontal axis

Rotate $y = \sqrt{2x + 3}$, $0 \le x \le 3$, about the x-axis. Find the curved surface area.

Solution. The radius is $y = \sqrt{2x + 3} > 0$. The given interval is the full required part of the curve, so the integration bounds remain 0 and 3.

$$y' = \frac{1}{\sqrt{2x + 3}}, \qquad \mathrm{d}s = \sqrt{\frac{2x + 4}{2x + 3}}\,\mathrm{d}x.$$

Multiplying radius and length element before integrating simplifies the expression:

$$S = 2\pi \int_0^3 \sqrt{2x + 3}\sqrt{\frac{2x + 4}{2x + 3}}\,\mathrm{d}x = 2\pi \int_0^3 \sqrt{2x + 4}\,\mathrm{d}x.$$

Set $u = 2x + 4$, so $\mathrm{d}x = \mathrm{d}u/2$ and the bounds become $4, 10$:

$$S = \pi \int_4^{10} u^{1/2}\,\mathrm{d}u = \boxed{\frac{2\pi}{3}\left(10\sqrt{10} - 8\right)}.$$

The radius is the distance to the horizontal axis; it is not x. The surface area uses the sloping element $\mathrm{d}s$, not just $\mathrm{d}x$.

Example 6. Surface area about the vertical axis

Rotate the curve $y = \dfrac{2}{3}x^{3/2}$, $0 \le x \le 1$, about the y-axis. Find the curved surface area.

Solution. A point (x, y) is a distance x from the y-axis. We may still integrate in x even though the rotation axis is vertical:

$$\rho = x, \qquad y' = \sqrt{x}, \qquad S = 2\pi \int_0^1 x\sqrt{1 + x}\,\mathrm{d}x.$$

Let $u = 1 + x$. Then $x = u - 1$, $\mathrm{d}u = \mathrm{d}x$, and u runs from 1 to 2.

$$\begin{aligned} S &= 2\pi \int_1^2 (u - 1)u^{1/2}\,\mathrm{d}u \\ &= 2\pi \left[\frac{2}{5}u^{5/2} - \frac{2}{3}u^{3/2}\right]_1^2 \\ &= 2\pi \left(\frac{8\sqrt{2}}{5} - \frac{4\sqrt{2}}{3} - \frac{2}{5} + \frac{2}{3}\right) = \boxed{\frac{8\pi}{15}(1 + \sqrt{2})}. \end{aligned}$$

Using y as the radius would describe rotation about the other axis and would give the wrong surface.

Volumes of solids of revolution

Disks and washers

Take a slice *perpendicular* to the rotation axis. If its rotation produces a washer with outer radius R and inner radius r, the cross-sectional area is $\pi(R^2 - r^2)$.

$$\text{Horizontal axis:} \quad V = \pi \int_a^b (R(x)^2 - r(x)^2)\,\mathrm{d}x,$$
$$\text{Vertical axis:} \quad V = \pi \int_c^d (R(y)^2 - r(y)^2)\,\mathrm{d}y.$$

A disk is the case $r = 0$. Radii are nonnegative distances from the axis. Square each radius separately; $(R - r)^2$ is not the area difference. Find the slice endpoints and any change in the presence of a hole before integrating.

Example 7. A region between an exponential curve and a line

The region $0 \le x \le \ln 2$, $1 \le y \le e^x$ is rotated about the x-axis. Find the volume.

Solution. A vertical slice runs from $y = 1$ to $y = e^x$. On rotation it gives a washer with $R = e^x$ and $r = 1$. The given vertical boundaries are $x = 0$ and $x = \ln 2$; at the first boundary the two curves meet, so the washer area is zero there.

$$\begin{aligned} V &= \pi \int_0^{\ln 2} ((e^x)^2 - 1^2)\,\mathrm{d}x \\ &= \pi \left[\frac{1}{2}e^{2x} - x\right]_0^{\ln 2} \\ &= \pi \left(2 - \ln 2 - \frac{1}{2}\right) = \boxed{\pi\left(\frac{3}{2} - \ln 2\right)}. \end{aligned}$$

The gap $e^x - 1$ is the thickness of the washer. It is not the radius of a disk.

Example 8. A hole appears partway through the solid

Rotate the region between $y = x^2$ and $y = 2x + 3$ about the y-axis. Find its volume using disks and washers.

Solution.

Find the region and its total vertical range

The curves meet where $x^2 = 2x + 3$, so

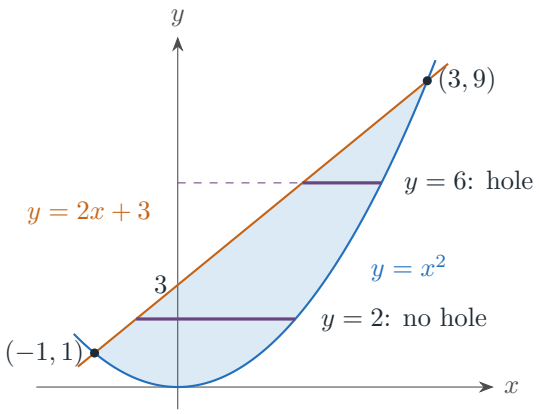
$$(x - 3)(x + 1) = 0, \qquad (x, y) = (-1, 1), (3, 9).$$

The parabola is below the line for $-1 \le x \le 3$. Its lowest point in the region is $(0, 0)$, so the region spans $0 \le y \le 9$.

A horizontal slice must satisfy both

$$-\sqrt{y} \le x \le \sqrt{y}, \qquad x \ge \frac{y - 3}{2}.$$

Thus its right endpoint is always $\sqrt{y}$. For $0 \le y \le 3$, the slice contains $x = 0$. Rotating it therefore fills a disk of radius $\sqrt{y}$, even if the left endpoint changes formula. For $3 < y \le 9$, the entire slice is to the right of the axis, so it makes a washer with inner radius $(y - 3)/2$.



Integrate the cross-sectional areas

$$\begin{aligned} V &= \pi \int_0^3 y \, dy + \pi \int_3^9 \left[y - \left(\frac{y-3}{2} \right)^2 \right] dy \\ &= \pi \int_0^9 y \, dy - \frac{\pi}{4} \int_3^9 (y-3)^2 \, dy \\ &= \frac{81\pi}{2} - \frac{\pi}{4} \left[\frac{(y-3)^3}{3} \right]_3^9 \\ &= \frac{81\pi}{2} - 18\pi = \boxed{\frac{45\pi}{2}}. \end{aligned}$$

The split occurs at $y = 3$ because this is where the inner radius first becomes positive. The intersection level $y = 1$ changes the left boundary of the planar slice, but does not create a hole in the rotated disk.

Cartesian double integrals: area and volume

A small rectangle and an iterated integral

Partition a planar region R into small rectangles. The limiting sum of $f(x, y)\Delta A$ is the double integral $\iint_R f \, dA$, when that limit exists. Continuity on a closed bounded region with ordinary piecewise smooth boundaries is a sufficient setting for the examples here.
On a rectangle $a \le x \le b$, $c \le y \le d$, a continuous integrand may be integrated in either order:

$$\iint_R f \, dA = \int_a^b \int_c^d f(x, y) \, dy \, dx = \int_c^d \int_a^b f(x, y) \, dx \, dy.$$

For a region between two graphs,

$$\begin{aligned} a \le x \le b, \quad g_1(x) \le y \le g_2(x) : &\quad \int_a^b \int_{g_1(x)}^{g_2(x)} f \, dy \, dx; \\ c \le y \le d, \quad h_1(y) \le x \le h_2(y) : &\quad \int_c^d \int_{h_1(y)}^{h_2(y)} f \, dx \, dy. \end{aligned}$$

The inner bounds are the entry and exit points of one slice. The outer bounds describe all positions of that slice. An inner bound cannot depend on the variable it is itself bounding.

Geometric interpretations

$$A(R) = \iint_R 1 \, dA, \qquad V = \iint_R (z_{\text{top}} - z_{\text{bottom}}) \, dA.$$

For volume, first check $z_{\text{top}} \ge z_{\text{bottom}}$ on the entire base. A double integral of a function that changes sign is a signed total; it is not automatically a geometric volume.

Example 9. Signed integral versus geometric volume

Over the rectangle $0 \le x \le 3$, $0 \le y \le 2$, evaluate $\iint_R (x-1)\,dA$. Then find the geometric volume between the plane $z = x - 1$ and the plane $z = 0$, with projection R.

Solution. The y-bounds are constant and the integrand does not depend on y:

$$\int_0^3 \int_0^2 (x-1)\,dy\,dx = \int_0^3 \big[(x-1)y\big]_0^2\,dx = 2\left[\frac{x^2}{2} - x\right]_0^3 = \boxed{3}.$$

For geometric volume, height is the distance between the planes, $|x-1|$. The plane changes from below to above $z = 0$ at $x = 1$, so split there:

$$\begin{aligned} V &= \int_0^1 \int_0^2 (1-x)\,dy\,dx + \int_1^3 \int_0^2 (x-1)\,dy\,dx \\ &= 2\left[x - \frac{x^2}{2}\right]_0^1 + 2\left[\frac{x^2}{2} - x\right]_1^3 \\ &= 1 + 4 = \boxed{5}. \end{aligned}$$

The signed integral subtracts the portion below the plane: $4 - 1 = 3$. The geometric volume adds both portions: $4 + 1 = 5$.

Example 10. Volume between two nonzero surfaces

Find the volume between $z = 6 - x^2 - y^2$ and $z = 1$ over $0 \le x \le 1$, $0 \le y \le 2$.

Solution. The vertical height is

$$h = (6 - x^2 - y^2) - 1 = 5 - x^2 - y^2.$$

On the rectangle, $x^2 + y^2 \le 1 + 4 = 5$, so $h \ge 0$. This confirms the stated upper and lower surfaces are in the correct order throughout the base.

$$\begin{aligned} V &= \int_0^1 \int_0^2 (5 - x^2 - y^2)\,dy\,dx \\ &= \int_0^1 \left[(5-x^2)y - \frac{y^3}{3}\right]_0^2 dx \\ &= \int_0^1 \left(\frac{22}{3} - 2x^2\right) dx \\ &= \left[\frac{22x}{3} - \frac{2x^3}{3}\right]_0^1 = \boxed{\frac{20}{3}}. \end{aligned}$$

Integrating $6 - x^2 - y^2$ alone would include the extra rectangular block between $z = 0$ and $z = 1$.

Example 11. A material layer with thickness measured in millimetres

A flat triangular part has vertices $(0,0)$, $(4,0)$ and $(0,2)$, with x, y measured in centimetres. A material layer has thickness

$$t(x,y) = 0.2 + 0.05x + 0.1y \quad \text{millimetres.}$$

Find its volume in cubic centimetres. Treat the layer as vertical above the flat part.

Solution.

Find the base and use consistent units

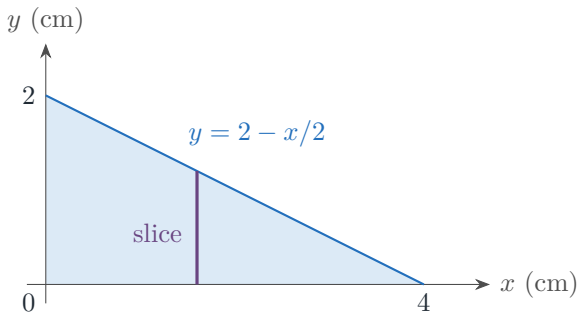
The sloping edge joins $(4,0)$ and $(0,2)$. Its equation is $x/4 + y/2 = 1$, or $y = 2 - x/2$. A vertical slice therefore has

$$0 \le x \le 4, \qquad 0 \le y \le 2 - \frac{x}{2}.$$

The area element $\mathrm{d}y\,\mathrm{d}x$ is in cm^2, so thickness must be in cm. Since 10 mm = 1 cm,

$$h(x,y) = 0.02 + 0.005x + 0.01y = \frac{1}{50} + \frac{x}{200} + \frac{y}{100} \quad \text{cm}.$$

It is positive on the whole triangle.



Integrate thickness over the base

Let $H = 2 - x/2$ for the inner substitution only. Then

$$\begin{aligned} V &= \int_0^4 \int_0^{2-x/2} \left(\frac{1}{50} + \frac{x}{200} + \frac{y}{100}\right) \mathrm{d}y\,\mathrm{d}x \\ &= \int_0^4 \left[\left(\frac{1}{50} + \frac{x}{200}\right) H + \frac{H^2}{200}\right] \mathrm{d}x. \end{aligned}$$

Here $(1/50 + x/200)H = 1/25 - x^2/400$ and $H^2/200 = 1/50 - x/100 + x^2/800$. Thus

$$\begin{aligned} V &= \int_0^4 \left(\frac{3}{50} - \frac{x}{100} - \frac{x^2}{800}\right) \mathrm{d}x \\ &= \left[\frac{3x}{50} - \frac{x^2}{200} - \frac{x^3}{2400}\right]_0^4 = \boxed{\frac{2}{15}\ \mathrm{cm}^3}. \end{aligned}$$

The base area is 4 cm^2, so the average thickness is $(2/15)/4 = 1/30$ cm, or $1/3$ mm. This lies between the vertex thicknesses $0.2, 0.4, 0.4$ mm, as expected.

Integrals and volumes over rectangles

Example 12. A linear integrand and a plane over a rectangle

(a) Evaluate $\displaystyle\int_0^1 \int_0^2 (x+y)\,\mathrm{d}y\,\mathrm{d}x$.

(b) Find the volume below $z = 4 - x - y$ and above the rectangle $0 \le x \le 1,\ 0 \le y \le 2$ in the xy-plane. Set up and evaluate a double integral.

Solution.

(a) Integrate with respect to the inner variable first

The inner variable is y, so x is constant during the first integration:

$$\begin{aligned} I &= \int_0^1 \left[xy + \frac{y^2}{2}\right]_{y=0}^{y=2} \mathrm{d}x = \int_0^1 (2x+2)\,\mathrm{d}x \\ &= \left[x^2 + 2x\right]_0^1 = 3. \end{aligned}$$

(b) Integrate the height above the base

The floor is $z = 0$, and the height is $4 - x - y$. Its smallest value on the rectangle is $4 - 1 - 2 = 1 > 0$, so the entire plane lies above the base. Thus

$$\begin{aligned} V &= \int_0^1 \int_0^2 (4 - x - y)\,\mathrm{d}y\,\mathrm{d}x \\ &= \int_0^1 \left[(4-x)y - \frac{y^2}{2}\right]_0^2 \mathrm{d}x = \int_0^1 (6 - 2x)\,\mathrm{d}x \\ &= \left[6x - x^2\right]_0^1 = 5. \end{aligned}$$

$\boxed{I = 3, \qquad V = 5 \text{ cubic units.}}$

Example 13. Two different rectangular bases

(a) Evaluate $\displaystyle\int_0^2 \int_0^1 (2x + y)\,\mathrm{d}y\,\mathrm{d}x$.

(b) Find the volume below $z = 3 + x + 2y$ and above $0 \le x \le 1,\ 0 \le y \le 1$. Set up and evaluate a double integral.

Solution.

(a) Use the limits in the stated integral

Here $0 \le x \le 2$ and $0 \le y \le 1$. Therefore

$$\begin{aligned} I &= \int_0^2 \left[2xy + \frac{y^2}{2}\right]_0^1 \mathrm{d}x = \int_0^2 \left(2x + \frac{1}{2}\right)\,\mathrm{d}x \\ &= \left[x^2 + \frac{x}{2}\right]_0^2 = 4 + 1 = 5. \end{aligned}$$

(b) Read the new base separately

The volume problem uses the unit square: both variables run from 0 to 1. The height $3 + x + 2y$ is at least 3 there. Hence

$$\begin{aligned} V &= \int_0^1 \int_0^1 (3 + x + 2y)\,\mathrm{d}y\,\mathrm{d}x \\ &= \int_0^1 \left[(3+x)y + y^2\right]_0^1 \mathrm{d}x = \int_0^1 (4 + x)\,\mathrm{d}x \\ &= \left[4x + \frac{x^2}{2}\right]_0^1 = \frac{9}{2}. \end{aligned}$$

$\boxed{I = 5, \qquad V = \dfrac{9}{2} \text{ cubic units.}}$

Example 14. A weighted linear integrand and a sloping plane

(a) Evaluate $\displaystyle\int_0^1 \int_0^2 (x+2y)\,dy\,dx$.

(b) Find the volume below $z = 5 - 2x - y$ and above $0 \le x \le 1,\ 0 \le y \le 1$. Set up and evaluate a double integral.

Solution.

(a) Evaluate the iterated integral

$$\begin{aligned} I &= \int_0^1 \Big[xy + y^2\Big]_0^2 \,dx = \int_0^1 (2x+4)\,dx \\ &= \Big[x^2 + 4x\Big]_0^1 = 5. \end{aligned}$$

(b) Use the plane's height over the unit square

The height is $5 - 2x - y \ge 5 - 2 - 1 = 2$, so it is nonnegative over the whole base.

$$\begin{aligned} V &= \int_0^1 \int_0^1 (5 - 2x - y)\,dy\,dx \\ &= \int_0^1 \left[(5-2x)y - \frac{y^2}{2}\right]_0^1 dx = \int_0^1 \left(\frac{9}{2} - 2x\right) dx \\ &= \left[\frac{9}{2}x - x^2\right]_0^1 = \frac{7}{2}. \end{aligned}$$

$$\boxed{I = 5, \qquad V = \frac{7}{2} \text{ cubic units.}}$$

Describing a curved region in both orders

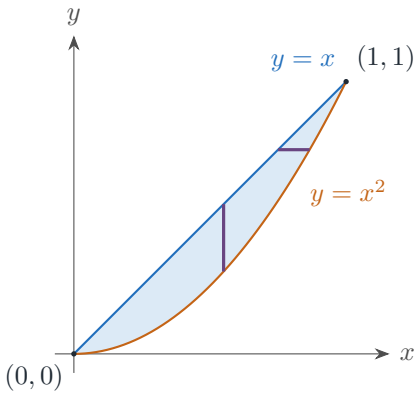
Example 15. A line and a parabola

Let D be the region bounded by $y = x$ and $y = x^2$. Write its limits in both orders, $dy\,dx$ and $dx\,dy$. Evaluate $\iint_D xy\,dA$ in both orders and verify that the values agree.

Solution.

Find the intersections and the two kinds of slice

The equation $x = x^2$ gives $x(x-1) = 0$, so the curves meet at $(0,0)$ and $(1,1)$. For $0 \le x \le 1$, $x^2 \le x$; the parabola is below the line.



A vertical slice gives

$$0 \le x \le 1, \qquad x^2 \le y \le x.$$

For a horizontal slice, $y = x$ becomes $x = y$, and the right branch of $y = x^2$ becomes $x = \sqrt{y}$. Hence

$$0 \le y \le 1, \qquad y \le x \le \sqrt{y}.$$

Evaluate in the order $\mathrm{d}y\,\mathrm{d}x$

$$\begin{aligned} I &= \int_0^1 \int_{x^2}^{x} xy \,\mathrm{d}y\,\mathrm{d}x = \int_0^1 \frac{x}{2}\left(x^2 - x^4\right)\,\mathrm{d}x \\ &= \frac{1}{2}\int_0^1 (x^3 - x^5)\,\mathrm{d}x = \frac{1}{2}\left[\frac{x^4}{4} - \frac{x^6}{6}\right]_0^1 = \frac{1}{24}. \end{aligned}$$

Evaluate in the order $\mathrm{d}x\,\mathrm{d}y$

$$\begin{aligned} I &= \int_0^1 \int_{y}^{\sqrt{y}} xy \,\mathrm{d}x\,\mathrm{d}y = \int_0^1 \frac{y}{2}\left(y - y^2\right)\,\mathrm{d}y \\ &= \frac{1}{2}\int_0^1 (y^2 - y^3)\,\mathrm{d}y = \frac{1}{2}\left[\frac{y^3}{3} - \frac{y^4}{4}\right]_0^1 = \frac{1}{24}. \end{aligned}$$

Both descriptions cover the same region once, and the two values agree.

$$\boxed{\iint_D xy \,\mathrm{d}A = \frac{1}{24}.}$$

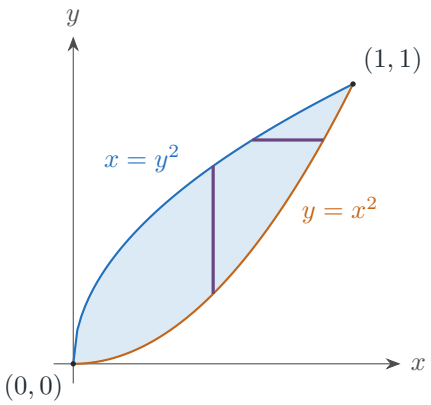
Example 16. Two parabolas with the coordinates interchanged

Let D be bounded by $y = x^2$ and $x = y^2$. Sketch D, write its limits in both orders, and evaluate $\iint_D xy \,\mathrm{d}A$.

Solution.

Locate the bounded region

Substitution gives $x = (x^2)^2 = x^4$, so $x = 0$ or $x = 1$. The intersections are $(0, 0)$ and $(1, 1)$. Both curves have nonnegative coordinates on the bounded loop. There, $x = y^2$ gives the upper graph $y = \sqrt{x}$.



The limits are

$$\begin{aligned} \mathrm{d}y\,\mathrm{d}x: &\quad 0 \le x \le 1, \quad x^2 \le y \le \sqrt{x}, \\ \mathrm{d}x\,\mathrm{d}y: &\quad 0 \le y \le 1, \quad y^2 \le x \le \sqrt{y}. \end{aligned}$$

Integrate across a vertical slice

$$I = \int_0^1 \int_{x^2}^{\sqrt{x}} xy \, dy \, dx = \int_0^1 \frac{x}{2}\left(x - x^4\right) dx$$
$$= \frac{1}{2}\int_0^1 (x^2 - x^5)\, dx = \frac{1}{2}\left[\frac{x^3}{3} - \frac{x^6}{6}\right]_0^1 = \frac{1}{12}.$$

Check. The other order gives $\displaystyle\int_0^1 \int_{y^2}^{\sqrt{y}} xy \, dx \, dy = \frac{1}{2}\int_0^1 (y^2 - y^5)\, dy = \frac{1}{12}$.

$$\boxed{\iint_D xy \, dA = \frac{1}{12}.}$$

Areas of general plane regions

Example 17. Two parabolas opening to the right

Draw the region enclosed by $x = y^2 - 1$ and $x = 2y^2 - 2$. Write its area as an iterated double integral and evaluate it.

Solution.

Use horizontal slices

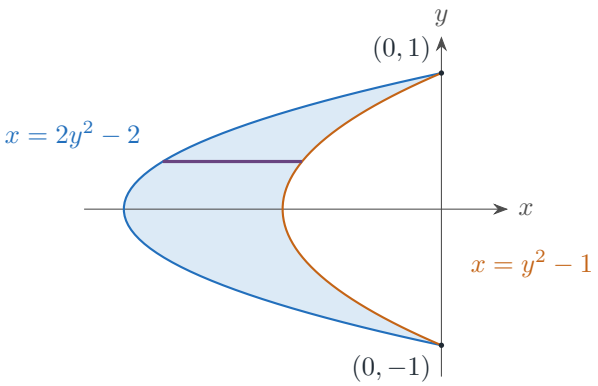
Both equations already give x in terms of y. At an intersection,

$$y^2 - 1 = 2y^2 - 2 \quad \Longrightarrow \quad y^2 = 1.$$

Thus the curves meet at $(0, -1)$ and $(0, 1)$. For $-1 \le y \le 1$,

$$(y^2 - 1) - (2y^2 - 2) = 1 - y^2 \ge 0.$$

So $x = 2y^2 - 2$ is the left boundary and $x = y^2 - 1$ is the right boundary.



Integrate the area element

$$A = \int_{-1}^{1} \int_{2y^2-2}^{y^2-1} 1 \, dx \, dy = \int_{-1}^{1} \left[(y^2 - 1) - (2y^2 - 2)\right] dy$$
$$= \int_{-1}^{1} (1 - y^2)\, dy = \left[y - \frac{y^3}{3}\right]_{-1}^{1} = \frac{2}{3} - \left(-\frac{2}{3}\right) = \frac{4}{3}.$$

$$\boxed{A = \frac{4}{3} \text{ square units.}}$$

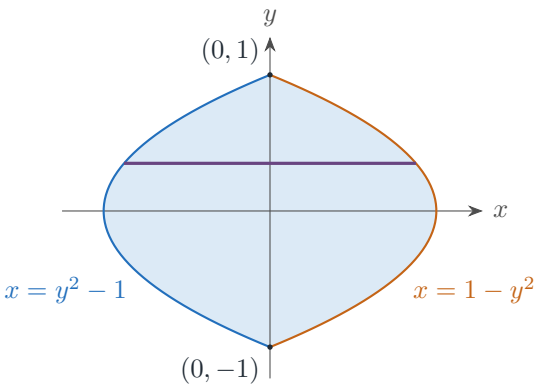
Example 18. Oppositely opening parabolas

Draw the region enclosed by $x = 1 - y^2$ and $x = y^2 - 1$. Write its area as an iterated double integral and evaluate it.

Solution. The intersections satisfy

$$1 - y^2 = y^2 - 1 \quad \Longrightarrow \quad y = \pm 1, \quad x = 0.$$

Between these values, the left boundary is $x = y^2 - 1$ and the right boundary is $x = 1 - y^2$. Their separation is $2 - 2y^2 \geq 0$.



Therefore

$$A = \int_{-1}^{1} \int_{y^2-1}^{1-y^2} 1 \, dx \, dy = \int_{-1}^{1} (2 - 2y^2) \, dy$$
$$= \left[2y - \frac{2y^3}{3} \right]_{-1}^{1} = \frac{4}{3} - \left(-\frac{4}{3} \right) = \frac{8}{3}.$$

$A = \dfrac{8}{3}$ square units.

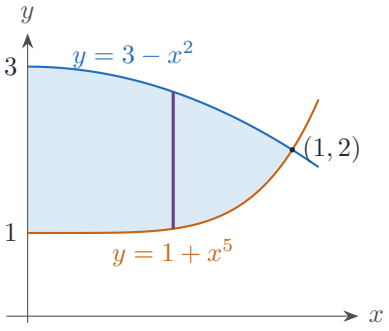
Example 19. A fifth-degree curve below a parabola

Find the area of the region bounded by $x = 0$, $y = x^5 + 1$, and $y = 3 - x^2$, using a double integral.

Solution.

Find where the right-hand boundaries meet

The intersection satisfies $x^5 + x^2 = 2$. On $x \geq 0$, the left-hand side increases, and its value at $x = 1$ is 2. Thus the right endpoint is $(1, 2)$. For $0 \leq x \leq 1$, $1 + x^5$ is the lower curve and $3 - x^2$ is the upper curve.



Integrate from the lower curve to the upper curve

$$A = \int_0^1 \int_{1+x^5}^{3-x^2} 1 \, dy \, dx = \int_0^1 \left[(3 - x^2) - (1 + x^5) \right] dx$$
$$= \int_0^1 (2 - x^2 - x^5) \, dx = \left[2x - \frac{x^3}{3} - \frac{x^6}{6} \right]_0^1 = \frac{3}{2}.$$

$A = \dfrac{3}{2}$ square units.

Example 20. A cubic curve below a line

Find the area of the region bounded by $x = 0$, $y = x^3 + 1$, and $y = 3 - x$, using a double integral.

Solution. The two graphs meet when

$$x^3 + 1 = 3 - x \quad \Longrightarrow \quad x^3 + x - 2 = (x - 1)(x^2 + x + 2) = 0.$$

The quadratic has no real root because its discriminant is $1 - 8 = -7$. Thus the meeting point is $(1, 2)$. The left boundary is $x = 0$, and on $0 \le x \le 1$ the cubic is below the line. The bounds are

$$0 \le x \le 1, \qquad 1 + x^3 \le y \le 3 - x.$$

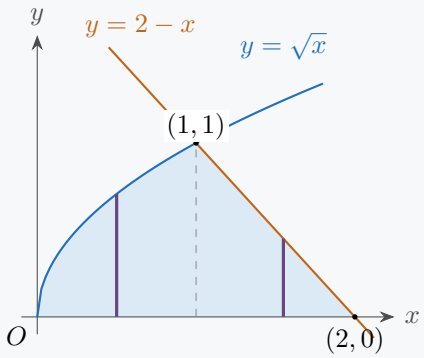
Hence

$$A = \int_0^1 \int_{1+x^3}^{3-x} 1 \, dy \, dx = \int_0^1 (2 - x - x^3) \, dx$$
$$= \left[2x - \frac{x^2}{2} - \frac{x^4}{4} \right]_0^1 = 2 - \frac{1}{2} - \frac{1}{4} = \frac{5}{4}.$$

$A = \dfrac{5}{4}$ square units.

Example 21. An area that splits in one order

The region shown is bounded by $y = \sqrt{x}$, $y = 2 - x$, and the x-axis. Write its area as iterated integral(s) in both orders and evaluate it using either order.



Solution.

Locate the change of upper boundary

At the joining point, $\sqrt{x} = 2 - x$. Since the square root is nonnegative, $x \le 2$. Squaring gives $(x-1)(x-4) = 0$; only $x = 1$ satisfies the original equation. The joining point is $(1, 1)$.

Horizontal slices: one integral

The lowest and highest y-values are 0 and 1. At a fixed y, the slice starts at $x = y^2$ and ends at $x = 2 - y$:

$$A = \int_0^1 \int_{y^2}^{2-y} 1 \, \mathrm{d}x \, \mathrm{d}y.$$

Vertical slices: two integrals

The lower boundary is always $y = 0$, but the upper boundary is $\sqrt{x}$ for $0 \le x \le 1$ and $2 - x$ for $1 \le x \le 2$. Therefore

$$A = \int_0^1 \int_0^{\sqrt{x}} 1 \, \mathrm{d}y \, \mathrm{d}x + \int_1^2 \int_0^{2-x} 1 \, \mathrm{d}y \, \mathrm{d}x.$$

Using horizontal slices,

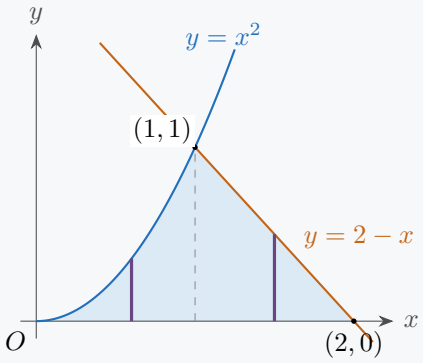
$$A = \int_0^1 (2 - y - y^2) \, \mathrm{d}y = \left[2y - \frac{y^2}{2} - \frac{y^3}{3} \right]_0^1 = \frac{7}{6}.$$

Check. The split order gives $\int_0^1 \sqrt{x} \, \mathrm{d}x + \int_1^2 (2 - x) \, \mathrm{d}x = \frac{2}{3} + \frac{1}{2} = \frac{7}{6}$.

$A = \dfrac{7}{6}$ square units.

Example 22. A parabola, a line and the axis

The shaded first-quadrant region is bounded by $y = x^2$, $y = 2 - x$, and the x-axis. Write its area as iterated integral(s) in both orders and evaluate it using either order.



Solution. The line and parabola meet when $x^2 = 2 - x$, so $(x + 2)(x - 1) = 0$. The shown region uses $(1, 1)$. The other intersection, $(-2, 4)$, lies outside it.

Write both descriptions of the shaded region

For horizontal slices, $0 \le y \le 1$. The left edge is the positive branch $x = \sqrt{y}$, and the right edge is $x = 2 - y$:

$$A = \int_0^1 \int_{\sqrt{y}}^{2-y} 1 \, dx \, dy.$$

For vertical slices, the upper edge changes at $x = 1$:

$$A = \int_0^1 \int_0^{x^2} 1 \, dy \, dx + \int_1^2 \int_0^{2-x} 1 \, dy \, dx.$$

Evaluate

$$\begin{aligned} A &= \int_0^1 (2 - y - \sqrt{y}) \, dy \\ &= \left[2y - \frac{y^2}{2} - \frac{2y^{3/2}}{3} \right]_0^1 = 2 - \frac{1}{2} - \frac{2}{3} = \frac{5}{6}. \end{aligned}$$

Check. The vertical slices give $\int_0^1 x^2 \, dx + \int_1^2 (2 - x) \, dx = \frac{1}{3} + \frac{1}{2} = \frac{5}{6}$.

$$\boxed{A = \frac{5}{6} \text{ square units.}}$$

Area and volume in simple applications

Example 23. A component with a triangular base

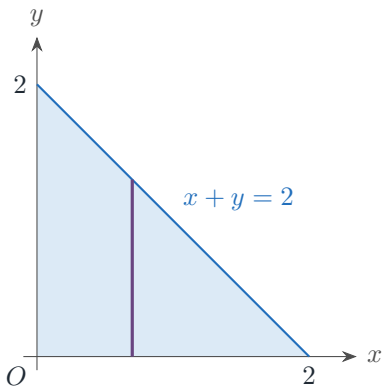
Let R be the first-quadrant triangle bounded by $x = 0$, $y = 0$, and $x + y = 2$.

(a) Find its area using a double integral.

(b) A solid engineering component has this base in the xy-plane and height $h(x, y) = 3 + x$ cm. Find its volume using a double integral.

Take the base coordinates in centimetres.

Solution. The sloping boundary is $y = 2 - x$, meeting the axes at $(2, 0)$ and $(0, 2)$. A vertical slice gives $0 \le x \le 2$, $0 \le y \le 2 - x$.



(a) Area

$$A = \int_0^2 \int_0^{2-x} 1 \, dy \, dx = \int_0^2 (2 - x) \, dx = \left[2x - \frac{x^2}{2} \right]_0^2 = 2 \text{ cm}^2.$$

(b) Volume

The height $3 + x$ is nonnegative, and a small base area dA supports a small volume $(3 + x) \, dA$. Thus

$$V = \int_0^2 \int_0^{2-x} (3 + x) \, dy \, dx = \int_0^2 (3 + x)(2 - x) \, dx$$
$$= \int_0^2 (6 - x - x^2) \, dx = \left[6x - \frac{x^2}{2} - \frac{x^3}{3} \right]_0^2 = \frac{22}{3} \text{ cm}^3.$$

$$A = 2 \text{ cm}^2, \qquad V = \frac{22}{3} \text{ cm}^3.$$

Example 24. A material layer over a curved base

Let R be bounded by $y = x^2$ and $y = x$.

(a) Find its area using a double integral.

(b) A material layer covers this base. Its thickness, measured perpendicular to the xy-plane, is $h(x, y) = 2 + x$ cm. Find the volume of material required.

Take the base coordinates in centimetres.

Solution.

Find the base limits

The boundaries meet when $x^2 = x$, at $x = 0, 1$. Between them, the lower curve is $y = x^2$ and the upper curve is $y = x$. Thus $0 \le x \le 1$, $x^2 \le y \le x$.

(a) Area

$$A = \int_0^1 \int_{x^2}^{x} 1 \, dy \, dx = \int_0^1 (x - x^2) \, dx = \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 = \frac{1}{6} \text{ cm}^2.$$

(b) Volume of the layer

The thickness is at least 2 cm. Integrating it over the same base gives

$$V = \int_0^1 \int_{x^2}^{x} (2 + x) \, dy \, dx = \int_0^1 (2 + x)(x - x^2) \, dx$$
$$= \int_0^1 (2x - x^2 - x^3) \, dx = \left[x^2 - \frac{x^3}{3} - \frac{x^4}{4} \right]_0^1 = \frac{5}{12} \text{ cm}^3.$$

$$A = \frac{1}{6} \text{ cm}^2, \qquad V = \frac{5}{12} \text{ cm}^3.$$

Example 25. A cover over a sideways parabola

Let R be bounded by $x = y^2$ and $x = 1$.

(a) Find its area using a double integral.

(b) A solid cover has this base in the xy-plane. Its height above the base is $h(x, y) = 2 + y$ cm. Find its volume using a double integral.

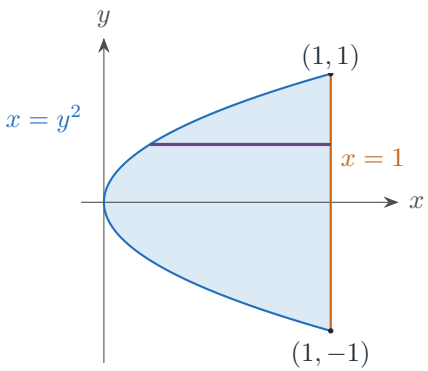
Take the base coordinates in centimetres.

Solution.

Include the whole bounded region

At $x = 1$, the parabola gives $y^2 = 1$, so both $y = -1$ and $y = 1$ are endpoints. A horizontal slice starts at $x = y^2$ and ends at $x = 1$:

$$-1 \le y \le 1, \qquad y^2 \le x \le 1.$$



(a) Area

$$A = \int_{-1}^{1} \int_{y^2}^{1} 1 \, dx \, dy = \int_{-1}^{1} (1 - y^2) \, dy = \left[y - \frac{y^3}{3} \right]_{-1}^{1} = \frac{4}{3} \text{ cm}^2.$$

(b) Volume

The height ranges from 1 to 3 cm, so it is positive over the entire base.

$$\begin{aligned} V &= \int_{-1}^{1}\int_{y^2}^{1}(2+y)\,dx\,dy = \int_{-1}^{1}(2+y)(1-y^2)\,dy \\ &= \int_{-1}^{1}(2+y-2y^2-y^3)\,dy \\ &= \left[2y+\frac{y^2}{2}-\frac{2y^3}{3}-\frac{y^4}{4}\right]_{-1}^{1} = \frac{8}{3}\ \text{cm}^3. \end{aligned}$$

Check. The odd terms $y - y^3$ cancel over $[-1, 1]$. The constant 2 cm part of the height contributes $2 \cdot \frac{4}{3} = \frac{8}{3}$ cm^3.

$$\boxed{A = \frac{4}{3}\ \text{cm}^2, \qquad V = \frac{8}{3}\ \text{cm}^3.}$$

Area and volume over the same base

Example 26. A downward parabola and a line

Let D be enclosed by $y = 2 - x^2$ and $y = x$.

(a) Find its area using a double integral.

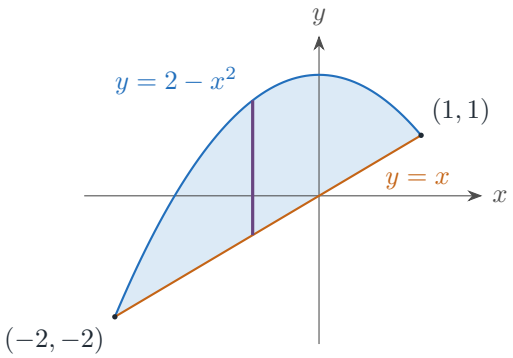
(b) Find the volume below $z = x^2$ and above D in the xy-plane.

Solution.

Use one description of the base for both parts

Solving $2 - x^2 = x$ gives $(x + 2)(x - 1) = 0$, hence $x = -2, 1$. On this interval the parabola is above the line, so

$$-2 \le x \le 1, \qquad x \le y \le 2 - x^2.$$



(a) Area: integrate 1

$$\begin{aligned} A &= \int_{-2}^{1}\int_{x}^{2-x^2} 1\,dy\,dx = \int_{-2}^{1}(2-x^2-x)\,dx \\ &= \left[2x-\frac{x^3}{3}-\frac{x^2}{2}\right]_{-2}^{1} = \frac{9}{2}. \end{aligned}$$

(b) Volume: integrate the height x^2

Although part of the base has $x < 0$, its height x^2 is always nonnegative.

$$\begin{aligned} V &= \int_{-2}^{1}\int_{x}^{2-x^2} x^2 \,dy\,dx = \int_{-2}^{1} x^2(2 - x^2 - x)\,dx \\ &= \int_{-2}^{1} (2x^2 - x^4 - x^3)\,dx \\ &= \left[\frac{2x^3}{3} - \frac{x^5}{5} - \frac{x^4}{4}\right]_{-2}^{1} = \frac{13}{60} - \left(-\frac{44}{15}\right) = \frac{63}{20}. \end{aligned}$$

$A = \dfrac{9}{2}$ square units, $\qquad V = \dfrac{63}{20}$ cubic units.

Example 27. A larger parabola-line region

Let D be enclosed by $y = 6 - x^2$ and $y = x$. Find its area using a double integral, then find the volume below $z = x^2$ and above D.

Solution.

Derive the limits

The boundary intersections satisfy

$$6 - x^2 = x \quad \Longrightarrow \quad (x + 3)(x - 2) = 0.$$

Thus the left endpoint has $x = -3$, the right endpoint has $x = 2$, and a vertical slice runs from the line $y = x$ to the parabola $y = 6 - x^2$:

$$-3 \le x \le 2, \qquad x \le y \le 6 - x^2.$$

Area

$$\begin{aligned} A &= \int_{-3}^{2}\int_{x}^{6-x^2} 1 \,dy\,dx = \int_{-3}^{2} (6 - x^2 - x)\,dx \\ &= \left[6x - \frac{x^3}{3} - \frac{x^2}{2}\right]_{-3}^{2} = \frac{22}{3} - \left(-\frac{27}{2}\right) = \frac{125}{6}. \end{aligned}$$

Volume

The height is $x^2 \ge 0$, so

$$\begin{aligned} V &= \int_{-3}^{2}\int_{x}^{6-x^2} x^2 \,dy\,dx = \int_{-3}^{2} (6x^2 - x^4 - x^3)\,dx \\ &= \left[2x^3 - \frac{x^5}{5} - \frac{x^4}{4}\right]_{-3}^{2} = \frac{28}{5} - \left(-\frac{513}{20}\right) = \frac{125}{4}. \end{aligned}$$

$A = \dfrac{125}{6}$ square units, $\qquad V = \dfrac{125}{4}$ cubic units.

Volumes under planes and curved surfaces

Example 28. A sloping plane over a parabola-line base

A solid has its base in the xy-plane between $y = 2 - x^2$ and $y = x$. Its upper surface is $z = x + 3$. Determine its volume.

Solution. The curves meet where $2 - x^2 = x$, giving $x = -2, 1$. Between these endpoints, $x \le y \le 2 - x^2$. The height from the xy-plane to the roof is $x + 3$, which is at least 1. The inner integration multiplies this height by the length of the base slice; the remaining integration adds the slice volumes across the base. Therefore

$$\begin{aligned} V &= \int_{-2}^{1} \int_{x}^{2-x^2} (x+3)\,dy\,dx \\ &= \int_{-2}^{1} (x+3)(2-x^2-x)\,dx \\ &= \int_{-2}^{1} (-x^3 - 4x^2 - x + 6)\,dx \\ &= \left[-\frac{x^4}{4} - \frac{4x^3}{3} - \frac{x^2}{2} + 6x \right]_{-2}^{1} \\ &= \frac{47}{12} - \left(-\frac{22}{3} \right) = \frac{45}{4}. \end{aligned}$$

$V = \dfrac{45}{4}$ cubic units.

Example 29. A plane meeting the base at one endpoint

A solid has its base in the xy-plane between $y = 4 - x^2$ and $y = 3x$. Its upper surface is $z = x + 4$. Determine its volume.

Solution.

Find the base and check the height

The intersections satisfy

$$4 - x^2 = 3x \quad \Longrightarrow \quad (x+4)(x-1) = 0,$$

so $-4 \le x \le 1$. For each such x, the lower base boundary is $3x$ and the upper boundary is $4 - x^2$. The roof height $x + 4$ is nonnegative; it is zero only at the left endpoint.

Set up and simplify the volume

$$\begin{aligned} V &= \int_{-4}^{1} \int_{3x}^{4-x^2} (x+4)\,dy\,dx \\ &= \int_{-4}^{1} (x+4)(4-x^2-3x)\,dx. \end{aligned}$$

Since $4 - x^2 - 3x = (x+4)(1-x)$, a short substitution avoids a longer polynomial expansion. Put $t = x + 4$, so $1 - x = 5 - t$; the bounds become $t = 0, 5$.

$$\begin{aligned} V &= \int_{0}^{5} t^2(5-t)\,dt = \int_{0}^{5} (5t^2 - t^3)\,dt \\ &= \left[\frac{5t^3}{3} - \frac{t^4}{4} \right]_{0}^{5} = \frac{625}{3} - \frac{625}{4} = \frac{625}{12}. \end{aligned}$$

$V = \dfrac{625}{12}$ cubic units.

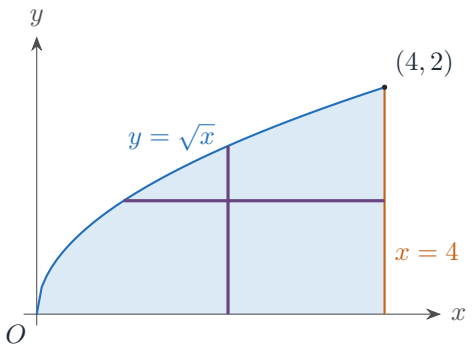
Example 30. Volume over a square-root base in both orders

A solid lies below $z = xy$ and above the region bounded by $y = \sqrt{x}$, $y = 0$, and $x = 4$. Sketch the base, set up the volume in both orders, and evaluate it.

Solution.

Read the base from vertical and horizontal slices

The square-root curve meets $y = 0$ at $(0, 0)$ and $x = 4$ at $(4, 2)$.



Vertical slices give $0 \le x \le 4$, $0 \le y \le \sqrt{x}$. For horizontal slices, $0 \le y \le 2$, and $y \le \sqrt{x}$ becomes $x \ge y^2$. Thus $y^2 \le x \le 4$.

Set up both orders and evaluate

The height xy is nonnegative throughout this first-quadrant base:

$$V = \int_0^4 \int_0^{\sqrt{x}} xy \, dy \, dx = \int_0^2 \int_{y^2}^4 xy \, dx \, dy.$$

Using the first order,

$$V = \int_0^4 \left[\frac{xy^2}{2} \right]_0^{\sqrt{x}} dx = \frac{1}{2} \int_0^4 x^2 \, dx = \left[\frac{x^3}{6} \right]_0^4 = \frac{32}{3}.$$

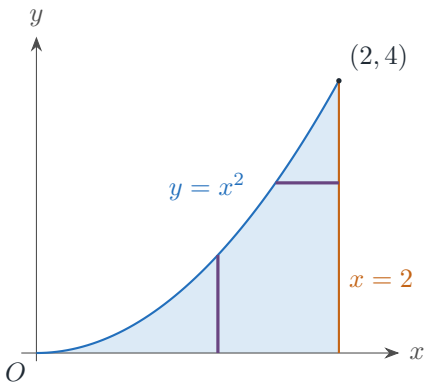
Check. The other order gives $\frac{1}{2} \int_0^2 (16y - y^5) \, dy = \left[4y^2 - \frac{y^6}{12} \right]_0^2 = \frac{32}{3}$.

$V = \frac{32}{3}$ cubic units.

Example 31. Volume over a parabolic base in both orders

A solid lies below $z = xy$ and above the region bounded by $y = x^2$, $y = 0$, and $x = 2$. Sketch the base, set up the volume in both orders, and evaluate it.

Solution. The bounded base lies below the parabola between $(0, 0)$ and $(2, 4)$. A vertical slice gives $0 \le x \le 2$, $0 \le y \le x^2$. For horizontal slices, $0 \le y \le 4$; on the positive branch, $y \le x^2$ gives $x \ge \sqrt{y}$, so $\sqrt{y} \le x \le 2$.



Since $xy \geq 0$ on the base,

$$V = \int_0^2 \int_0^{x^2} xy \, dy \, dx = \int_0^4 \int_{\sqrt{y}}^2 xy \, dx \, dy.$$

The first order gives

$$V = \int_0^2 \left[\frac{xy^2}{2}\right]_0^{x^2} dx = \frac{1}{2}\int_0^2 x^5 \, dx = \left[\frac{x^6}{12}\right]_0^2 = \frac{16}{3}.$$

Check. The other order gives $\frac{1}{2}\int_0^4 (4y - y^2)\, dy = \left[y^2 - \frac{y^3}{6}\right]_0^4 = \frac{16}{3}$.

$V = \frac{16}{3}$ cubic units.

Example 32. A solid below a circular cylinder

A solid in the first octant is bounded by the cylinder $y^2 + z^2 = 4$ and the planes $x = 2y$, $x = 0$, and $z = 0$. Sketch its base in the xy-plane, set up its volume as a double integral, and evaluate it.

Solution.

Obtain the finite base from all the bounding surfaces

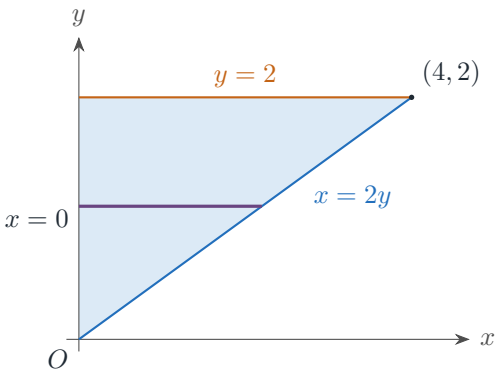
The first octant means $x, y, z \geq 0$. The cylinder gives the upper surface

$$z = \sqrt{4 - y^2}.$$

For this height to be real, $0 \leq y \leq 2$. The side planes then give $0 \leq x \leq 2y$. Thus the projection is the triangle

$$D = \{(x, y) : 0 \leq y \leq 2,\ 0 \leq x \leq 2y\}.$$

The bound $y = 2$ comes from the cylinder meeting $z = 0$. The two side planes by themselves would leave an unbounded region in the xy-plane.



Choose the simpler inner integral

The height depends only on y, so integrate with respect to x first:

$$V = \int_0^2 \int_0^{2y} \sqrt{4-y^2}\,\mathrm{d}x\,\mathrm{d}y$$
$$= \int_0^2 \left[x\sqrt{4-y^2}\right]_0^{2y} \mathrm{d}y = 2\int_0^2 y\sqrt{4-y^2}\,\mathrm{d}y.$$

Let $u = 4 - y^2$, so $\mathrm{d}u = -2y\,\mathrm{d}y$. As y goes from 0 to 2, u goes from 4 to 0. Therefore

$$V = -\int_4^0 u^{1/2}\,\mathrm{d}u = \int_0^4 u^{1/2}\,\mathrm{d}u = \left[\frac{2u^{3/2}}{3}\right]_0^4 = \frac{16}{3}.$$

$V = \dfrac{16}{3}$ cubic units.

Example 33. A cylinder with a different axis and radius

A solid in the first octant is bounded by $x^2 + z^2 = 9$, $y = x$, $y = 0$, and $z = 0$. Sketch its base in the xy-plane, set up its volume as a double integral, and evaluate it.

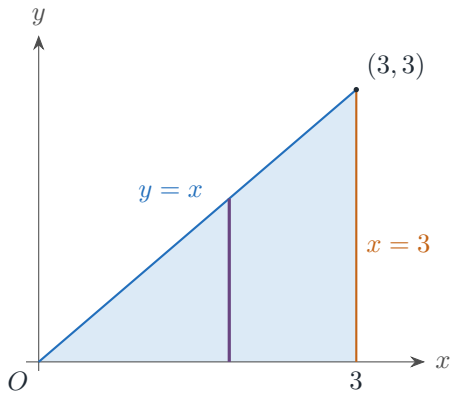
Solution.

Project the solid onto the base plane

Since $z \geq 0$, the upper surface is $z = \sqrt{9-x^2}$. It is real for $0 \leq x \leq 3$ in the first octant. Between the side planes, $0 \leq y \leq x$. Hence

$$D = \{(x, y) : 0 \leq x \leq 3,\ 0 \leq y \leq x\}.$$

The edge $x = 3$ comes from the cylinder meeting the base $z = 0$; the side planes alone do not supply this finite endpoint.



Integrate the height

The height depends only on x, so the inner y-integration is immediate:

$$V = \int_0^3 \int_0^x \sqrt{9-x^2}\,\mathrm{d}y\,\mathrm{d}x = \int_0^3 x\sqrt{9-x^2}\,\mathrm{d}x.$$

Put $u = 9 - x^2$, so $\mathrm{d}u = -2x\,\mathrm{d}x$. The transformed bounds are 9 and 0:

$$V = -\frac{1}{2}\int_9^0 u^{1/2}\,\mathrm{d}u = \frac{1}{2}\left[\frac{2u^{3/2}}{3}\right]_0^9 = \frac{9^{3/2}}{3} = 9.$$

$V = 9$ cubic units.

Choosing the order that simplifies the calculation

Example 34. An exponential integrand on a square

Evaluate $\iint_R xe^{xy}\,\mathrm{d}A$, where $R=[0,1]\times[0,1]$. Choose the simpler order of integration and explain your choice.

Solution. For fixed x, the chain rule gives

$$\frac{\partial}{\partial y}e^{xy}=xe^{xy}.$$

Thus $\mathrm{d}y\,\mathrm{d}x$ is convenient: the first integration exactly matches this derivative. The rectangle supplies constant limits for both variables:

$$\begin{aligned}I&=\int_0^1\int_0^1 xe^{xy}\,\mathrm{d}y\,\mathrm{d}x=\int_0^1\left[e^{xy}\right]_{y=0}^{y=1}\,\mathrm{d}x\\&=\int_0^1(e^x-1)\,\mathrm{d}x=\left[e^x-x\right]_0^1=(e-1)-1=e-2.\end{aligned}$$

$\boxed{I=e-2.}$

Example 35. A trigonometric integrand on a rectangle

Evaluate $\iint_R y\cos(xy)\,\mathrm{d}A$, where $R=[0,1]\times[0,\pi/2]$. Choose the simpler order and explain why it is convenient.

Solution. The first interval describes x, and the second describes y. For fixed y,

$$\frac{\partial}{\partial x}\sin(xy)=y\cos(xy).$$

Therefore integrate with respect to x first:

$$\begin{aligned}I&=\int_0^{\pi/2}\int_0^1 y\cos(xy)\,\mathrm{d}x\,\mathrm{d}y\\&=\int_0^{\pi/2}\left[\sin(xy)\right]_{x=0}^{x=1}\,\mathrm{d}y=\int_0^{\pi/2}\sin y\,\mathrm{d}y\\&=\left[-\cos y\right]_0^{\pi/2}=0-(-1)=1.\end{aligned}$$

$\boxed{I=1.}$

Reversing the order of integration

Keep the region fixed; describe new slices

For a continuous function on the bounded regions used here, either order gives the same double integral provided both sets of limits describe the same region exactly once.

1. Read the original limits as inequalities and sketch the region.
2. Find the full range of the new outer variable.
3. Hold that outer variable fixed. Follow a slice in the direction of the new inner variable; its entry and exit points give the inner lower and upper limits.
4. If the entry or exit boundary changes as the slice moves, split the outer interval at that value and add the resulting integrals.

The function being integrated stays the same. The new limits and differentials must match the new order; simply interchanging the printed bounds does not describe the region correctly.

Example 36. An exponential curve as an upper boundary

Sketch the integration region, reverse the order, and evaluate

$$\int_0^1 \int_1^{e^x} \mathrm{d}y\,\mathrm{d}x.$$

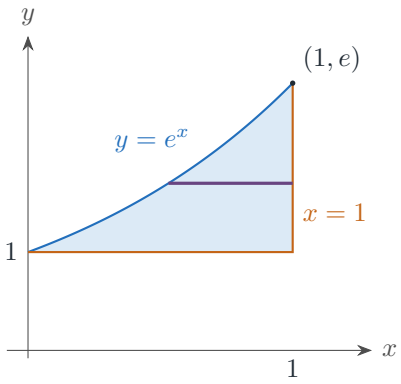
Solution.

Read the original bounds as inequalities

The integrand is 1. The bounds describe

$$0 \le x \le 1, \qquad 1 \le y \le e^x.$$

The region lies above $y = 1$, below $y = e^x$, and to the left of $x = 1$.



Describe a horizontal slice

The lowest y-value is 1, and the highest is e. Taking logarithms of $y \le e^x$ gives $\ln y \le x$. Therefore the new limits are

$$1 \le y \le e, \qquad \ln y \le x \le 1.$$

Evaluate the reversed integral

$$I = \int_1^e \int_{\ln y}^1 1\,\mathrm{d}x\,\mathrm{d}y = \int_1^e (1 - \ln y)\,\mathrm{d}y.$$

Integration by parts gives $\int \ln y\,\mathrm{d}y = y\ln y - \int y(1/y)\,\mathrm{d}y = y\ln y - y$. Hence

$$\boxed{I = [2y - y\ln y]_1^e = e - 2.}$$

Example 37. An exponential curve as a left boundary

Sketch the region, reverse the order, and evaluate

$$\int_0^{\ln 2} \int_{e^y}^{2} \mathrm{d}x\,\mathrm{d}y.$$

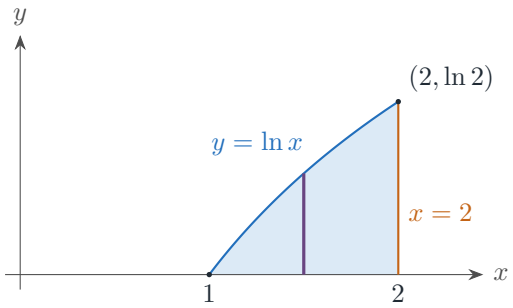
Solution.

Turn the bounds into a region

The integrand is 1, and

$$0 \le y \le \ln 2, \qquad e^y \le x \le 2.$$

The curve $x = e^y$ can also be written $y = \ln x$. It meets $y = 0$ at $(1, 0)$ and $x = 2$ at $(2, \ln 2)$.



Use vertical slices

The horizontal range is $1 \le x \le 2$. Since $e^y \le x$ means $y \le \ln x$, the vertical slice runs from $y = 0$ to $y = \ln x$. Therefore

$$I = \int_1^2 \int_0^{\ln x} 1\,\mathrm{d}y\,\mathrm{d}x = \int_1^2 \ln x\,\mathrm{d}x.$$

Integration by parts, with $u = \ln x$ and $\mathrm{d}v = \mathrm{d}x$, gives $\int \ln x\,\mathrm{d}x = x\ln x - x$. Hence

$$I = [x\ln x - x]_1^2 = (2\ln 2 - 2) - (-1) = 2\ln 2 - 1.$$

$$\boxed{\int_1^2 \int_0^{\ln x} \mathrm{d}y\,\mathrm{d}x = 2\ln 2 - 1.}$$

Example 38. Reversal creates a useful factor of x

Draw the integration region, reverse the order, and hence evaluate

$$\int_0^{2\sqrt{\ln 3}} \int_{y/2}^{\sqrt{\ln 3}} e^{x^2}\,\mathrm{d}x\,\mathrm{d}y.$$

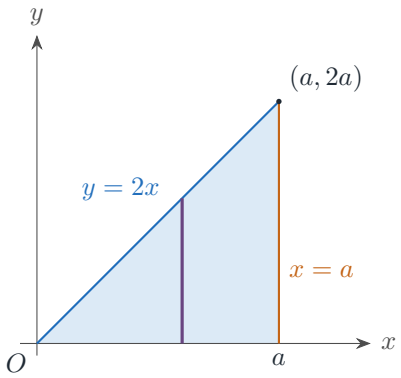
Solution.

Describe the triangular region

Write $a = \sqrt{\ln 3} > 0$. The bounds say

$$0 \le y \le 2a, \qquad y/2 \le x \le a.$$

The boundary $x = y/2$ is the line $y = 2x$, and $x = a$ is the right edge.



For a vertical slice, x runs from 0 to a, while y runs from 0 to $2x$. Thus

$$I = \int_0^a \int_0^{2x} e^{x^2}\,\mathrm{d}y\,\mathrm{d}x.$$

Evaluate after reversing

During the inner y-integration, e^{x^2} is constant:

$$I = \int_0^a \left[ye^{x^2}\right]_0^{2x}\,\mathrm{d}x = 2\int_0^a xe^{x^2}\,\mathrm{d}x.$$

Now $u = x^2$ gives $\mathrm{d}u = 2x\,\mathrm{d}x$, with $u = 0$ to $u = \ln 3$:

$$I = \int_0^{\ln 3} e^u\,\mathrm{d}u = [e^u]_0^{\ln 3} = 3 - 1 = 2.$$

The original inner integral $\int e^{x^2}\,\mathrm{d}x$ has no elementary antiderivative. Reversing the order supplies the factor $2x$ needed for the substitution.

$\boxed{I = 2.}$

Example 39. A steeper triangular region

Draw the integration region, reverse the order, and hence evaluate

$$\int_0^{3\sqrt{\ln 2}} \int_{y/3}^{\sqrt{\ln 2}} e^{x^2}\,\mathrm{d}x\,\mathrm{d}y.$$

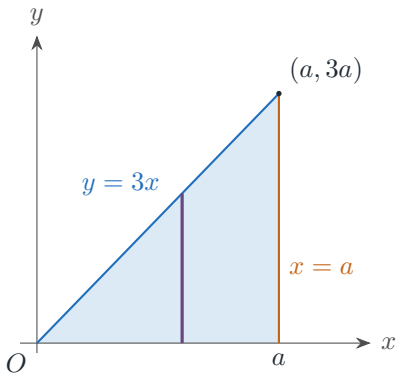
Solution.

Find the new limits from the same region

Let $a = \sqrt{\ln 2} > 0$. The original limits are

$$0 \le y \le 3a, \qquad y/3 \le x \le a.$$

The sloping edge is $y = 3x$; the other edges are $y = 0$ and $x = a$.



A vertical slice therefore gives $0 \le x \le a$, $0 \le y \le 3x$.

Reverse and evaluate

$$\begin{aligned} I &= \int_0^a \int_0^{3x} e^{x^2}\,\mathrm{d}y\,\mathrm{d}x = \int_0^a \left[ye^{x^2}\right]_0^{3x}\,\mathrm{d}x \\ &= 3\int_0^a xe^{x^2}\,\mathrm{d}x. \end{aligned}$$

With $u = x^2$, $x\,\mathrm{d}x = \frac{1}{2}\,\mathrm{d}u$, and the new upper limit is $a^2 = \ln 2$. Thus

$$I = \frac{3}{2}\int_0^{\ln 2} e^u\,\mathrm{d}u = \frac{3}{2}\left[e^u\right]_0^{\ln 2} = \frac{3}{2}(2-1) = \frac{3}{2}.$$

$$\boxed{I = \frac{3}{2}.}$$

Polar double integrals

Change the coordinates, the bounds and the area element

Use standard polar coordinates measured from the original origin:

$$x = r\cos\theta, \qquad y = r\sin\theta, \qquad r \geq 0, \qquad x^2 + y^2 = r^2.$$

Angles are in radians. A small polar cell has radial width Δr and approximate arc width $r\Delta\theta$, so its area is approximately $r\Delta r\Delta\theta$. Equivalently,

$$\frac{\partial(x,y)}{\partial(r,\theta)} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} = r, \qquad \boxed{\mathrm{d}A = r\,\mathrm{d}r\,\mathrm{d}\theta}.$$

If a ray at angle θ enters at $r = a(\theta)$ and exits at $r = b(\theta)$, then

$$\iint_R f(x,y)\,\mathrm{d}A = \int_\alpha^\beta \int_{a(\theta)}^{b(\theta)} f(r\cos\theta, r\sin\theta)\, r\,\mathrm{d}r\,\mathrm{d}\theta.$$

Choose an angular interval that sweeps the region once. For continuous integrands on the bounded regions used here, the usual polar change of variables is valid; the origin and boundary overlaps have area zero.

Example 40. A half-annulus and a nonradial integrand

Evaluate $\iint_R x\,\mathrm{d}A$, where R is the right-hand half of the annulus $9 \leq x^2 + y^2 \leq 25$.

Solution. The two circles give $3 \leq r \leq 5$. The condition $x \geq 0$ means the ray points into the right half-plane, which is covered once by

$$-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}.$$

The integrand becomes $x = r\cos\theta$; the area element contributes a second factor r:

$$\begin{aligned} I &= \int_{-\pi/2}^{\pi/2} \int_3^5 r^2\cos\theta\,\mathrm{d}r\,\mathrm{d}\theta \\ &= \int_{-\pi/2}^{\pi/2} \frac{125 - 27}{3}\cos\theta\,\mathrm{d}\theta \\ &= \frac{98}{3}[\sin\theta]_{-\pi/2}^{\pi/2} = \boxed{\frac{196}{3}}. \end{aligned}$$

Symmetry about the x-axis does not cancel this integral: x has the same positive sign above and below that axis. A full annulus would have cancellation between its left and right halves.

Example 41. A circular cap with a nonzero radial entry point

Use polar coordinates to find the area of the region

$$x^2 + y^2 \leq 4, \qquad y \geq 1.$$

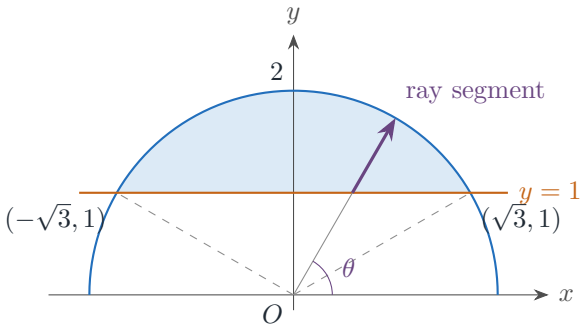
Solution. The circle has radius 2 and the horizontal line lies above the origin. A ray enters the region on $y = 1$ and leaves on the circle, so

$$r\sin\theta = 1 \implies r_{\text{entry}} = \csc\theta, \qquad r_{\text{exit}} = 2.$$

The radial interval exists only when $\csc\theta \leq 2$, or $\sin\theta \geq 1/2$. In the upper half-plane this gives

$$\boxed{\frac{\pi}{6} \leq \theta \leq \frac{5\pi}{6}, \qquad \csc\theta \leq r \leq 2.}$$

The endpoint rays pass through the intersections $(-\sqrt{3}, 1)$ and $(\sqrt{3}, 1)$.



Since this is area, the Cartesian integrand is 1, and the polar integrand including the area factor is r:

$$\begin{aligned} A &= \int_{\pi/6}^{5\pi/6} \int_{\csc\theta}^{2} r \, \mathrm{d}r \, \mathrm{d}\theta \\ &= \frac{1}{2} \int_{\pi/6}^{5\pi/6} (4 - \csc^2\theta) \, \mathrm{d}\theta \\ &= \left[2\theta + \frac{1}{2}\cot\theta \right]_{\pi/6}^{5\pi/6} = \boxed{\frac{4\pi}{3} - \sqrt{3}}. \end{aligned}$$

Here $\frac{\mathrm{d}}{\mathrm{d}\theta}\cot\theta = -\csc^2\theta$, and $\cot(\pi/6) = \sqrt{3}$, $\cot(5\pi/6) = -\sqrt{3}$. Starting each ray at $r = 0$ would incorrectly include the triangular region below $y = 1$.

Example 42. Convert a polar integral back to Cartesian coordinates

Describe the region and evaluate

$$I = \int_{0}^{\pi/4} \int_{\sec\theta}^{2\sec\theta} r^2 \cos\theta \, \mathrm{d}r \, \mathrm{d}\theta$$

by converting it to a Cartesian double integral.

Solution.

Translate the region

The angle range gives the first-quadrant sector $0 \le y \le x$. Also $\cos\theta > 0$ on this range, so the radial inequalities can be multiplied by $\cos\theta$ without reversing them:

$$\sec\theta \le r \le 2\sec\theta \quad \Longleftrightarrow \quad 1 \le r\cos\theta \le 2 \quad \Longleftrightarrow \quad 1 \le x \le 2.$$

The region is therefore the trapezium with vertices $(1,0), (2,0), (2,2), (1,1)$. Vertical slices give

$$1 \le x \le 2, \qquad 0 \le y \le x.$$

Separate the area factor from the function

Write the given integrand as $(r\cos\theta)\, r \, \mathrm{d}r \, \mathrm{d}\theta$. Since $r\cos\theta = x$ and $r \, \mathrm{d}r \, \mathrm{d}\theta = \mathrm{d}A$, the Cartesian function is x, not $x\sqrt{x^2+y^2}$.

$$I = \int_{1}^{2} \int_{0}^{x} x \, \mathrm{d}y \, \mathrm{d}x = \int_{1}^{2} x^2 \, \mathrm{d}x = \left[\frac{x^3}{3} \right]_{1}^{2} = \boxed{\frac{7}{3}}.$$

As a direct polar check,

$$I = \frac{7}{3} \int_{0}^{\pi/4} \sec^2\theta \, \mathrm{d}\theta = \frac{7}{3}[\tan\theta]_{0}^{\pi/4} = \frac{7}{3}.$$

The extra r belongs to the polar area element; it must disappear when that area element is replaced by $\mathrm{d}x \, \mathrm{d}y$.

Example 43. Volume between a cone and a horizontal plane

Find the volume enclosed between the cone $z = \sqrt{x^2 + y^2}$ and the plane $z = 3$.

Solution. The surfaces meet when $\sqrt{x^2 + y^2} = 3$, giving the circle $x^2 + y^2 = 9$. A vertical segment has lower endpoint on the cone and upper endpoint on the plane, so the base is the disk $r \le 3$ and the height is $3 - r \ge 0$.

$$\begin{aligned} V &= \int_0^{2\pi} \int_0^3 (3 - r) r \, \mathrm{d}r \, \mathrm{d}\theta \\ &= \int_0^{2\pi} \left[\frac{3}{2} r^2 - \frac{1}{3} r^3 \right]_0^3 \mathrm{d}\theta \\ &= \int_0^{2\pi} \frac{9}{2} \, \mathrm{d}\theta = \boxed{9\pi}. \end{aligned}$$

The height $3 - r$ and the area factor r have different roles. The result agrees with the volume of a cone of base radius 3 and height 3: $\frac{1}{3}\pi(3)^2(3) = 9\pi$.

Cartesian triple integrals

Adding small volume elements

Divide a solid D into small rectangular cells. A cell has volume $\Delta V = \Delta x \Delta y \Delta z$. Adding weighted cell volumes and taking a limit defines

$$\iiint_D f(x, y, z) \, \mathrm{d}V = \lim \sum_i f(x_i, y_i, z_i) \Delta V_i,$$

when the limit exists as cell sizes tend to zero. A sufficient setting here is a continuous function on a closed bounded solid whose boundary consists of finitely many smooth surface pieces.
In Cartesian coordinates, $\mathrm{d}V = \mathrm{d}x \, \mathrm{d}y \, \mathrm{d}z$. For geometric volume, each cell contributes its own volume, so

$$\boxed{\mathrm{Vol}(D) = \iiint_D 1 \, \mathrm{d}V.}$$

A general triple integral weights the cells and need not be a volume.

Three integrations, performed from the inside out

For a rectangular box $[a, b] \times [c, d] \times [e, h]$, Fubini's theorem gives

$$\iiint_D f \, \mathrm{d}V = \int_a^b \int_c^d \int_e^h f(x, y, z) \, \mathrm{d}z \, \mathrm{d}y \, \mathrm{d}x.$$

First integrate in z while x, y stay fixed. Next integrate in y while x stays fixed. Finally integrate in x. For a continuous function on a box, any of the six variable orders is valid, with the corresponding bounds.

Example 44. A box and a weighted integral

Let D be the box $-1 \le x \le 1$, $2 \le y \le 4$, $0 \le z \le 2$. Find its volume and evaluate $\iiint_D (x+y+z)\,dV$.

Solution.

Volume: use integrand 1

$$\begin{aligned} V &= \int_{-1}^{1}\int_{2}^{4}\int_{0}^{2} 1\,dz\,dy\,dx \\ &= \int_{-1}^{1}\int_{2}^{4} [z]_0^2\,dy\,dx = \int_{-1}^{1} [2y]_2^4\,dx \\ &= \int_{-1}^{1} 4\,dx = [4x]_{-1}^{1} = \boxed{8}. \end{aligned}$$

The side lengths are $1-(-1)=2$, $4-2=2$ and $2-0=2$, so their product also gives 8.

Weighted integral: keep the remaining variables constant

While integrating in z, $x+y$ is a constant:

$$\begin{aligned} J &= \int_{-1}^{1}\int_{2}^{4} \left[(x+y)z + \frac{z^2}{2}\right]_0^2 dy\,dx \\ &= \int_{-1}^{1}\int_{2}^{4} (2x+2y+2)\,dy\,dx \\ &= \int_{-1}^{1} [2xy + y^2 + 2y]_2^4\,dx \\ &= \int_{-1}^{1} (4x+16)\,dx = [2x^2+16x]_{-1}^{1} = \boxed{32}. \end{aligned}$$

The contribution of x cancels on the symmetric interval $[-1,1]$, but the contributions of y and z do not. The value 32 is the weighted integral; the box volume remains 8.

Bounds for a solid between two surfaces

Suppose a vertical line above (x,y) enters the solid at $z=b(x,y)$ and leaves at $z=t(x,y)$. The set of all allowed (x,y) is its projection R on the xy-plane. If

$$a \le x \le b_0, \qquad g_1(x) \le y \le g_2(x),$$

then

$$\iiint_D f\,dV = \int_a^{b_0}\int_{g_1(x)}^{g_2(x)}\int_{b(x,y)}^{t(x,y)} f\,dz\,dy\,dx.$$

For the order $dz\,dy\,dx$, the z-bounds may depend on x, y; the y-bounds may depend on x; the outer x-bounds are constants. Bounds cannot contain the variable they bound or a variable already integrated out.

Example 45. A plane and the three coordinate planes

Find the volume in the first octant bounded by $2x+3y+z=6$ and the three coordinate planes.

Solution.

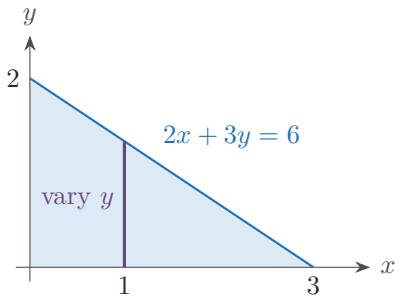
Obtain the projection, then the vertical bounds

First octant means $x, y, z \ge 0$. The lower surface is $z=0$ and the upper surface is $z = 6-2x-3y$. A nonnegative vertical segment exists precisely when

$$2x+3y \le 6.$$

Thus the projection is the triangle with vertices $(0,0),(3,0),(0,2)$. For a fixed x, the line $2x+3y=6$ gives $y=2-2x/3$, so

$$\boxed{0\le x\le 3,\quad 0\le y\le 2-\frac{2x}{3},\quad 0\le z\le 6-2x-3y.}$$



Projection on the xy-plane

At the sample base point $(1,1)$, the vertical line runs from $z=0$ to $z=1$. On the sloping edge $2x+3y=6$, the height is zero. These checks confirm the geometry of the bounds.

Carry out all three integrations

Put $H=2-2x/3$; then $6-2x=3H$. Integrating first in z produces the height:

$$\begin{aligned}
V &= \int_0^3\int_0^H\int_0^{6-2x-3y} 1\,\mathrm{d}z\,\mathrm{d}y\,\mathrm{d}x\\
&= \int_0^3\int_0^H (6-2x-3y)\,\mathrm{d}y\,\mathrm{d}x\\
&= \int_0^3 \left[(6-2x)y-\frac{3}{2}y^2\right]_0^H \mathrm{d}x\\
&= \int_0^3 \left(3H^2-\frac{3}{2}H^2\right)\mathrm{d}x\\
&= \frac{2}{3}\int_0^3 (3-x)^2\,\mathrm{d}x = \frac{2}{3}\left[-\frac{(3-x)^3}{3}\right]_0^3 = \boxed{6}.
\end{aligned}$$

As a geometric check, the solid is a pyramid with triangular base area $\frac{1}{2}(3)(2)=3$ and perpendicular height 6. Its volume is $\frac{1}{3}(3)(6)=6$.

Example 46. A curved lower surface determines the projection

A solid satisfies $0\le x\le 3$, $y\ge 0$, and lies between the lower surface $z=y^2$ and the upper surface $z=4-x$. Find its volume.

Solution.

Do not assume the base is a rectangle

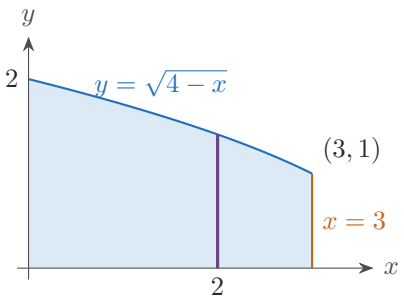
At an allowed base point, the lower surface cannot be higher than the upper surface:

$$y^2\le 4-x.$$

Since $y\ge 0$, this gives $0\le y\le\sqrt{4-x}$. The explicit side planes restrict x to $0\le x\le 3$, so the full description is

$$\boxed{0\le x\le 3,\quad 0\le y\le\sqrt{4-x},\quad y^2\le z\le 4-x.}$$

The upper bound on x is 3, even though the surfaces alone could meet as far as $x=4$. The stated side plane cuts the solid off earlier.



Projection on the xy-plane

For instance, in the vertical section $x = 2$, the solid lies between the parabola $z = y^2$ and the line $z = 2$, with $0 \le y \le \sqrt{2}$. The gap closes exactly at $y = \sqrt{2}$.

Integrate through the vertical gap

$$\begin{aligned} V &= \int_0^3 \int_0^{\sqrt{4-x}} \int_{y^2}^{4-x} 1 \, \mathrm{d}z \, \mathrm{d}y \, \mathrm{d}x \\ &= \int_0^3 \int_0^{\sqrt{4-x}} (4 - x - y^2) \, \mathrm{d}y \, \mathrm{d}x \\ &= \int_0^3 \left[(4-x)y - \frac{y^3}{3} \right]_0^{\sqrt{4-x}} \mathrm{d}x \\ &= \frac{2}{3} \int_0^3 (4-x)^{3/2} \, \mathrm{d}x. \end{aligned}$$

Let $u = 4 - x$. Then $\mathrm{d}u = -\mathrm{d}x$, and the bounds $x = 0, 3$ become $u = 4, 1$:

$$V = \frac{2}{3} \int_1^4 u^{3/2} \, \mathrm{d}u = \frac{4}{15} [u^{5/2}]_1^4 = \boxed{\frac{124}{15}}.$$

Why a triple volume integral becomes a double integral

For any solid lying between $z = b(x, y)$ and $z = t(x, y)$ over a suitable base R,

$$\iiint_D 1 \, \mathrm{d}V = \iint_R \left(\int_{b(x,y)}^{t(x,y)} 1 \, \mathrm{d}z \right) \mathrm{d}A = \iint_R (t(x, y) - b(x, y)) \, \mathrm{d}A.$$

The first integration produces the height. In the last example, that height is $4 - x - y^2$. Putting this height into the triple integrand as well would count it twice: volume uses integrand 1 before integrating in z.